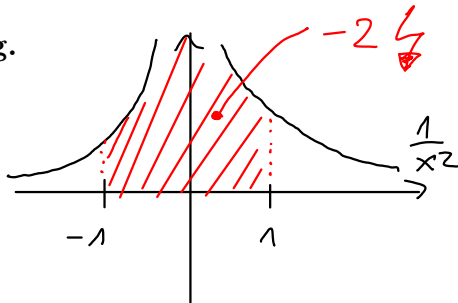


INTEGRALE

* **Fehlerteufel.** Wo steckt der Fehler? $\int_{-1}^1 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^1 = -\frac{1}{1} - \left(-\frac{1}{-1}\right) = -2$

Lösung.



Fehler: $\frac{1}{x^2}$ hat $D = \mathbb{R} \setminus \{0\}$.

Es wurde über die Definitionslücke $x=0$ integriert!

$$\int_{-1}^1 \frac{1}{x^2} dx \stackrel{\text{symmetrisch}}{=} 2 \cdot \int_0^1 \frac{1}{x^2} dx = 2 \cdot \lim_{\varepsilon \rightarrow 0^+} \underbrace{\int_{\varepsilon}^1 \frac{1}{x^2} dx}_{\left[-\frac{1}{x}\right]_{\varepsilon}^1} = \infty \text{ ist die Fläche!}$$

$$\underbrace{-1 + \frac{1}{\varepsilon}}_{\rightarrow \infty}$$

Eigener Lösungsversuch.

Uneigentliche Integrale. Berechnen Sie:

* 1. $\int_1^{\infty} \frac{2}{x^3} dx$

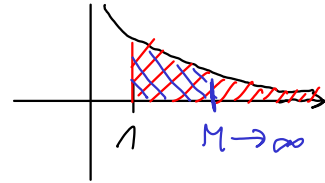
2. $\int_{-\infty}^0 e^x dx$

3. $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$ (Hinweis: Stammfkt. ist eine trigonometr. Fkt.)

Lösung.

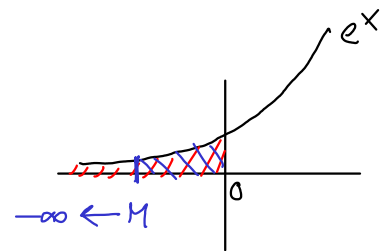
$$1. \int_1^{\infty} \frac{2}{x^3} dx = \lim_{M \rightarrow \infty} \int_1^M \frac{2}{x^3} dx = \lim_{M \rightarrow \infty} \left(-\frac{1}{M^2} + \frac{1}{1^2} \right) = 1$$

$$\int x^{-3} dx = \frac{x^{-2}}{-2} + C \rightarrow \left[-\frac{2}{2x^2} \right]_1^M$$



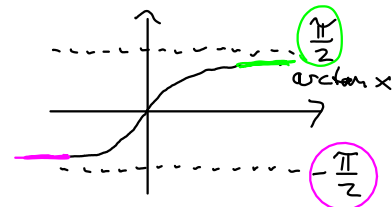
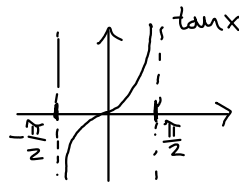
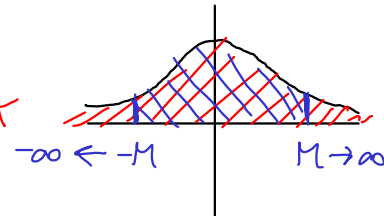
$$2. \int_{-\infty}^0 e^x dx = \lim_{M \rightarrow -\infty} \int_M^0 e^x dx = \lim_{M \rightarrow -\infty} \left(e^0 - \underbrace{e^M}_{\rightarrow 0} \right) = 1$$

$$\left[e^x \right]_M^0$$



$$3. \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \lim_{M \rightarrow \infty} \int_{-M}^M \frac{1}{1+x^2} dx = \lim_{M \rightarrow \infty} \left(\underbrace{\arctan(M)}_{\frac{\pi}{2}} - \underbrace{\arctan(-M)}_{-\frac{\pi}{2}} \right) = \pi$$

$$\left[\arctan x \right]_{-M}^M$$



Eigener Lösungsversuch.

Partielle Integration. Berechnen Sie die folgenden Integrale:

* 1. $\int x e^x dx$

3. $\int x^2 \ln x dx$

2. $\int e^x (2 - x^2) dx$

4. $\int_1^e \ln x dx$ Hinweis: $\ln x = 1 \cdot \ln x$

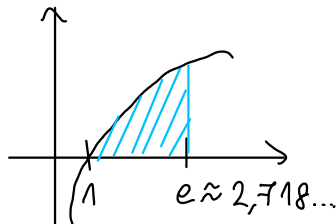
Lösung.

$$1. \int \underset{f}{x} \underset{g}{e^x} dx = \underset{f}{e^x} \cdot x - \int \underset{f}{e^x} \cdot \underset{g'}{1} dx = x e^x - e^x + C.$$

$$\begin{aligned} 2. \int \underset{f}{e^x} \underset{g}{(2-x^2)} dx &= \underset{f}{e^x} \underset{g}{(2-x^2)} - \int \underset{f}{e^x} \cdot \underset{g'}{(-2x)} dx = \underset{f}{e^x} \underset{g}{(2-x^2)} + \underbrace{\int \underset{f}{e^x} \cdot \underset{g}{2x} dx}_{\substack{e^x \cdot 2x - \int \underset{f}{e^x} \cdot \underset{g'}{2} dx}} \\ &= e^x (2-x^2) + e^x \cdot 2x - 2e^x + C \\ &= e^x (-x^2 + 2x) + C \end{aligned}$$

$$\begin{aligned} 3. \int \underset{f}{x^2} \underset{g}{\ln x} dx &= \frac{x^3}{3} \cdot \ln x - \int \frac{x^3}{3} \cdot \underset{g'}{\frac{1}{x}} dx = \frac{x^3}{3} \ln x - \frac{x^3}{9} + C \\ &= \frac{x^3}{3} \left(\ln x - \frac{1}{3} \right) + C \end{aligned}$$

$$\begin{aligned} 4. \int_1^e \underset{f}{1} \cdot \underset{g}{\ln x} dx &= \left[\underset{f}{x} \cdot \ln x \right]_1^e - \int_1^e \underset{f}{x} \cdot \underset{g'}{\frac{1}{x}} dx = \left[x \ln x - x \right]_1^e = \underbrace{(e \ln e - e)}_0 - \underbrace{(1 \cdot \ln 1 - 1)}_0 \\ &= \underline{\underline{1}}. \end{aligned}$$



Eigener Lösungsversuch.

Substitutionsregel. Berechnen Sie die folgenden Integrale:

* 1. $\int_0^{\pi/2} \sin(2x) dx$

2. $\int \sin(x) \cos^3(x) dx$

3. $\int \frac{x}{5+x^2} dx$

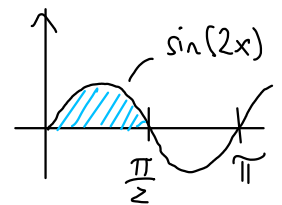
4. $\int_0^2 \frac{t^2}{1+t^3} dt$

5. $\int \frac{\arctan(z)}{1+z^2} dz$ Hinweis: $\frac{\arctan(z)}{1+z^2} = (\arctan(z))' \cdot \frac{1}{1+z^2}$
oder: $u = \arctan z$

Lösung.

1. $\frac{1}{2} \int_0^{\pi/2} \underbrace{\sin(2x)}_{f(u(x))} \cdot \underbrace{2}_{u'(x)} dx = \frac{1}{2} \cdot \left[\underbrace{-\cos(2x)}_{F(u(x))} \right]_0^{\pi/2} = \frac{1}{2} \cdot \left(\underbrace{-\cos(\pi)}_{-1} + \underbrace{\cos(0)}_1 \right) = 1$

ODER: $\int_0^{\pi/2} \sin(2x) dx = \int_{u=0}^{u=\pi/2} \sin(u) \frac{du}{2} = \left[-\frac{1}{2} \cos(u) \right]_0^{\pi/2} = \dots = 1.$



2. $-\int (\cos(x))^3 \cdot (-\sin(x)) dx = -\frac{(\cos(x))^4}{4} + C = -\frac{1}{4} \cos^4(x) + C$

ODER: $\int \sin(x) (\cos(x))^3 dx \stackrel{u=\cos(x)}{\stackrel{\frac{du}{dx}=-\sin(x)}{=}} \int \sin(x) \cdot u^3 \frac{du}{-\sin(x)} = -\int u^3 du = -\frac{u^4}{4} + C = -\frac{\cos^4(x)}{4} + C.$

3. $\frac{1}{2} \int \frac{1}{5+x^2} \cdot 2x dx = \frac{1}{2} \cdot \ln|5+x^2| + C$ ODER: $u=5+x^2, \frac{du}{dx}=2x \dots$

4. $\frac{1}{3} \int_0^2 \frac{1}{1+t^3} \cdot 3t^2 dt = \frac{1}{3} \cdot \left[\ln|1+t^3| \right]_0^2 = \frac{1}{3} \cdot \left(\underbrace{\ln|1+8|}_{\ln 9} - \underbrace{\ln|1|}_0 \right) = \frac{1}{3} \ln 9 = \frac{2}{3} \ln 3.$

3./4. mit logarithmischer Integration $\boxed{\int \frac{f'}{f} dx = \ln|f| + C}$

3. $\frac{1}{2} \int \frac{\overset{f'}{\cancel{2x}}}{\underset{f}{5+x^2}} dx = \frac{1}{2} \ln|5+x^2| + C$

4. $\frac{1}{3} \int_0^2 \frac{\overset{f'}{\cancel{3t^2}}}{\underset{f}{1+t^3}} dx = \frac{1}{3} \left[\ln|1+t^3| \right]_0^2 = \dots$

5. $\int (\arctan z)' \cdot \frac{1}{1+z^2} dz = \frac{(\arctan z)^2}{2} + C$

ODER: $\int \frac{\arctan z}{1+z^2} dz \stackrel{u=\arctan z}{\stackrel{\frac{du}{dz}=\frac{1}{1+z^2}}{=}} \int \frac{u}{\cancel{1+z^2}} \cdot \frac{\cancel{1}}{\cancel{1+z^2}} du = \int u du = \frac{u^2}{2} + C = \frac{(\arctan z)^2}{2} + C$

Eigener Lösungsversuch.

Partialbruchzerlegung. Integrieren Sie folgende rationale Funktionen:

* 1. $\int \frac{-2x^2 + x + 8}{x^3 - 4x^2 + 4x} dx$ *← NST suchen*

2. $\int \frac{9x^3 - 6x^2 + x}{(x+1)^3(x-3)^2} dx$

3. $\int \frac{2x^2 - 3x + 3}{(x-1)(x^2+1)} dx$

4. $\int \frac{2x^3 + 3x}{(x^2 - x + 2)^2} dx$ *← keine NST*
← keine NST

Lösung. Siehe dazu DorFuchs Mathe-Song auf YouTube zur PBZ:

1. NST Nenner: $x^3 - 4x^2 + 4x = x \cdot (x^2 - 4x + 4) = x \cdot (x-2)^2 \Rightarrow x_1=0, x_{2,3}=2$ doppelte NST!

PBZ: $\frac{-2x^2 + x + 8}{x(x-2)^2} \stackrel{\text{Ansatz}}{=} \frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2} = \frac{A \cdot (x-2)^2 + B \cdot x \cdot (x-2) + C \cdot x}{x \cdot (x-2)^2}$

zähler
 $-2x^2 + x + 8 \stackrel{!}{=} A(x-2)^2 + Bx(x-2) + Cx$

$$\left. \begin{array}{l} x=0 \Rightarrow 8=4A \\ x=2 \Rightarrow 2=2C \\ x=1 \Rightarrow 7=A-B+C \end{array} \right\} \begin{array}{l} A=2 \\ B=-4 \\ C=1 \end{array}$$

$$\frac{-2x^2 + x + 8}{x(x-2)^2} = \frac{2}{x} - \frac{4}{x-2} + \frac{1}{(x-2)^2}$$

$$\Rightarrow \int \frac{-2x^2 + x + 8}{x(x-2)^2} dx = 2 \int \frac{1}{x} dx - 4 \int \frac{1}{x-2} dx + \int \frac{1}{(x-2)^2} dx$$

subst.: $(x-2)^{-2} \cdot 1$ ODER: $u=x-2, \frac{du}{dx}=1$

$$= 2 \ln|x| - 4 \ln|x-2| + \frac{(x-2)^{-1}}{-1} + C = 2 \ln|x| - 4 \ln|x-2| - \frac{1}{x-2} + C$$

2. PBZ: $\frac{9x^3 - 6x^2 + x}{(x+1)^3(x-3)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} + \frac{D}{x-3} + \frac{E}{(x-3)^2} = \frac{2}{(x+1)^3(x-3)^2}$

3-fache NST 2-fache NST

$$9x^3 - 6x^2 + x = A(x+1)^2(x-3)^2 + B(x+1)(x-3)^2 + C(x-3)^2 + D(x+1)^3(x-3) + E(x+1)^3$$

$$\left. \begin{array}{lcl} x = -1 & \Rightarrow & -16 = 16C \\ x = 0 & \Rightarrow & 0 = 9A + 9B + 9C - 3D + E \\ x = 1 & \Rightarrow & 4 = 16A + 8B + 4C - 16D + 8E \\ x = 2 & \Rightarrow & 50 = 9A + 3B + C - 27D + 27E \\ x = 3 & \Rightarrow & 192 = 64E \end{array} \right\} \begin{array}{l} A = -1 \\ B = 2 \\ C = -1 \\ D = 1 \\ E = 3 \end{array}$$

$$\frac{9x^3 - 6x^2 + x}{(x+1)^3(x-3)^2} = -\frac{1}{x+1} + \frac{2}{(x+1)^2} - \frac{1}{(x+1)^3} + \frac{1}{x-3} + \frac{3}{(x-3)^2}$$

$$\int \frac{9x^3 - 6x^2 + x}{(x+1)^3(x-3)^2} dx = -\ln|x+1| + 2 \cdot \frac{(x+1)^{-1}}{-1} - \frac{(x+1)^{-2}}{-2} + \ln|x-3| + 3 \cdot \frac{(x-3)^{-1}}{-1} + C.$$

3. PBZ: $\frac{2x^2 - 3x + 3}{(x-1)(x^2+1)} \stackrel{\text{Ansatz PBZ bei quadratischen Faktoren}}{=} \frac{A}{x-1} + \frac{Bx+C}{x^2+1} = \frac{A \cdot (x^2+1) + (Bx+C)(x-1)}{(x-1)(x^2+1)}$ Vergleichen!
 $\rightarrow A, B, C$ berechnen!

$x=0$ einsetzen: $\underbrace{2 \cdot 0^2 - 3 \cdot 0 + 3}_0 = \underbrace{A \cdot (0^2+1)}_A + \underbrace{(B \cdot 0 + C)(0-1)}_{-C} \Rightarrow 3 = A - C \quad (\text{I})$

NST $x=1$ einsetzen: $2 = A \cdot 2 \Rightarrow \underline{A=1} \quad (\text{II})$

$x=2$ einsetzen: $5 = 5A + 2B + C \quad (\text{III})$

(II) in (I) : $3 = 1 - C \Rightarrow \underline{C = -2}$ in (III) : $5 = 5 \cdot 1 + 2B - 2 \Rightarrow \underline{B = 1}$

$$\int \frac{2x^2 - 3x + 3}{(x-1)(x^2+1)} dx = \underbrace{\int \frac{1}{x-1} dx}_{\ln|x-1| + C} + \underbrace{\int \frac{x-2}{x^2+1} dx}_{\frac{1}{2} \int \frac{2x}{x^2+1} dx - \int \frac{2}{x^2+1} dx} = \ln|x-1| + \frac{1}{2} \ln|x^2+1| - 2 \arctan(x) + C$$

$$\frac{1}{2} \ln|x^2+1| + C - 2 \arctan(x) + C$$

4. P82:
$$\frac{2x^3 + 3x}{(x^2 - x + 2)^2} = \frac{Ax + B}{x^2 - x + 2} + \frac{Cx + D}{(x^2 - x + 2)^2}$$

2-facher quadratischer Term!

$$2x^3 + 3x = (Ax + B)(x^2 - x + 2) + Cx + D$$

$$\left. \begin{array}{lcl} x = -1 & \Rightarrow & -5 = -4A + 4B - C + D \\ x = 0 & \Rightarrow & 0 = 2B + D \\ x = 1 & \Rightarrow & 5 = 2A + 2B + C + D \\ x = 2 & \Rightarrow & 22 = 8A + 4B + 2C + D \end{array} \right\} \begin{array}{l} A = 2 \\ B = 2 \\ C = 1 \\ D = -4 \end{array}$$

$$\frac{2x^3 + 3x}{(x^2 - x + 2)^2} = \frac{2x + 2}{x^2 - x + 2} + \frac{x - 4}{(x^2 - x + 2)^2}$$

Wie integriert man $\int \frac{2x+2}{x^2-x+2} dx$?
keine NST!

$$\int \frac{2x+2}{x^2-x+2} dx = \underbrace{\int \frac{2x-1}{x^2-x+2} dx}_{\ln|x^2-x+2| + C} + \int \frac{3}{x^2-x+2} dx$$

(*) $\frac{1}{(u(x))^2 + 1}$ (für $\arctan(u(x))' = \frac{1}{(u(x))^2 + 1} u'(x)$)

$$\begin{aligned} (*) : \frac{3}{x^2-x+2} &= \frac{3}{x^2 - 2 \cdot \frac{1}{2}x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + 2} = \frac{3}{\left(x - \frac{1}{2}\right)^2 + \frac{7}{4}} = \frac{3}{\frac{7}{4}} \cdot \frac{1}{\frac{4}{7}\left(x - \frac{1}{2}\right)^2 + 1} \\ &\quad \text{①. quadr. Ergänzen} \quad \text{②. ausklammern!} \\ &= \frac{12}{7} \cdot \frac{1}{\left(\sqrt{\frac{4}{7}} \cdot \left(x - \frac{1}{2}\right)\right)^2 + 1} \end{aligned}$$

$$\Rightarrow \int \frac{3}{x^2-x+2} dx = \frac{12}{7} \cdot \frac{1}{\sqrt{\frac{4}{7}}} \int \frac{1}{\left(\sqrt{\frac{4}{7}} \left(x - \frac{1}{2}\right)\right)^2 + 1} \sqrt{\frac{4}{7}} dx = \frac{12}{7} \cdot \sqrt{\frac{7}{4}} \cdot \arctan\left(\sqrt{\frac{4}{7}} \left(x - \frac{1}{2}\right)\right) + C$$

$\arctan\left(\sqrt{\frac{4}{7}} \left(x - \frac{1}{2}\right)\right) + C$

ODER: Subst. mit $u(x) = \sqrt{\frac{4}{7}} \left(x - \frac{1}{2}\right)$, $\frac{du}{dx} = \sqrt{\frac{4}{7}}$

Eigener Lösungsversuch.